Report spektroscopy I

Introduction

Experiment

Results

k_0 denotes the k-value which corresponds to the 0-th order. The measured value is $k_0 = 950$. This value is wrong. More details are in the discussion section.

the first measurement is to fit the wavelength of the middle pixel position in dependance of dial position (k-value). The results are the following:

wavelength [nm]	388.8	447.1	471.3	492.1	504.7	667.8	706.5
k-value	1310 ± 1	1131 ± 1	1101 ± 1	1072 ± 1	1070 ± 1	849 ± 1	800 ± 1

The first column contains the wavelength in nm of the measured light-strips. The value is the literature value. The second column contains the measurements. The measurement is the k-value. This is light from the He-Lamp.

the second measurement is for a fixed k-value and measures data to fit the wavelength in dependance of pixel position the results are the following:

wavelength [nm]	492.1	501.5	504.7
pixel position	475 ± 1	802 ± 1	920 ± 1

The first column is again the known wavelengths of the measured light-strips in nm. the second column is the pixel position on the image. the image is 1440px wide, therefore the pixel position p has $0 \leq p < 1440$

the third data-set is from the Hydrogen-Lamp to measure the wavelengths of the Balmer-Series.

n	3	4	5	6
k-value	860 ± 1	1080 ± 1	1112 ± 1	1120 ± 1

the first column is the Balmer-Series. The are the corresponding measurements of the k-values.

Data Analysis

The fit function used is of the form $\lambda(p, k) = bk + a + \frac{p - p_{\text{cent}}}{d}$ we used this function beacause our k_0 -value was wrong. Therefore we now use three fit-parameter instead of two like in the manualll. with the first data-set we obtained values for a and b. we used scipy.curve_fit for the regression. For the first data-set we used the fit $\lambda_c(k) = bk + a$ since $p = p_{\text{cent}}$. This gives the values $a = 1220 \pm 2, b = -0.663 \pm 0.002$ **insert plot**

for the second data-set the fit function is $\lambda(p, k_{\text{fix}}) = bk_{\text{fix}} + a + (p - \frac{p_{(\text{cent})}}{d}$. this is the final fit function but with a fixed $k = k_{\text{fix}}$ and a, b from above. this gives $d = 42.6 \pm 0.1$. **insert plot**

from the fit we calculate the lattice constant g as $g = \frac{1}{d} = (5.87 \pm 0.02) \cdot 10^{-5}$.

with the complete fit function we obtained the following wavelengths of the Balmer-Series through the fit function: $\lambda_3 = 650 \pm 2$ nm, $\lambda_4 = 504 \pm 2$ nm, $\lambda_5 = 483 \pm 2$ nm, $\lambda_{6_=} 477 \pm 2$ nm.

from the average of the obtained wavelengths we calculate the Rydberg-Frequency as $R_{\text{H}} = (3.07 \pm 0.01)e15$ therefore the limit of the Balmer-Series is $\lambda_{\text{min}} = 391 \pm 2$ nm. From the Rydberg-Frequency we continue to calculate the ionisation energy as $E_{\text{ion}} = \frac{hR_{\text{H}}}{e} = 12.7 \pm 0.1$ eV where h is Plancks constant and e is the proton charge.

Discussion

The k_0-value measured in the experiment is $k_0 = 950$. This measurment does not correspond to the actual k_0-vlaue but to the 587.5 nm wavelength light. This wavelength is yellow and was therefore misinterpreted as the 0-th order. In order to correct for this mistake, we used for the first fit-function $\lambda_c\ (k) = bk + a$ instead of $\lambda_c\ (k) = b(k - k_0)$. The latter function is used in the manual.

The measurements of the Balmer-Series have given:

literature value [nm]	656	486	434	410
experimental value [nm]	650 ± 2	504 ± 2	483 ± 2	477 ± 2

Also the experimental values from the Rydberg-Frequency, ionisation energy and the limit of the Balmer-Series deviate from the literature values.

	Rydberg-Frequency [Hz]	ionisation energy [eV]	limit of Series [nm]
literature value	3.29 e 15	13.6	365
experimental value	$(3.07 \pm 0.01)e15$	12.7 ± 0.1	391 ± 2

The errors from the Rydberg-Frequency, ionisation energy and limit of the Balmer-Series directly propagate from the errors in measurement of the H-Lines. The values obtained from the H-Lines come from the fit-function and the measured k-values for each H-Line. Both of them have their own errors, which contribute. The errors trace back to the originally set deviations of $\Delta k = \pm 1$ and $\Delta p = \pm 1$ where Δk is the error of the k-value and Δp the error in pixel position.

Conclusion